

EXPERIMENT 5

TITLE

Water Jug Problem

AIM

To write a Python program to solve the **Water Jug Problem** using state-space search.

OBJECTIVE

- To understand the Water Jug Problem.
- To generate valid water states.
- To obtain the required amount of water.
- To apply Artificial Intelligence search techniques.

ALGORITHM

Step 1: Define the capacities of both jugs.

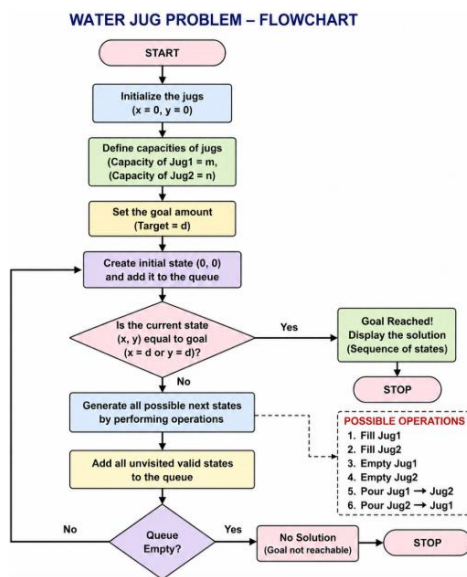
Step 2: Start with both jugs empty.

Step 3: Fill, empty, or transfer water between the jugs.

Step 4: Check whether the required quantity is obtained.

Step 5: Repeat until the goal state is reached.

FLOWCHART



PYTHON CODE

```
from collections import deque

def water_jug(jug1, jug2, target):

    visited = set()
    queue = deque()

    queue.append((0,0))

    while queue:

        x,y = queue.popleft()

        if (x,y) in visited:
            continue

        visited.add((x,y))

        print((x,y))

        if x==target or y==target:
            print("Goal Reached")
            return

        next_states = [
            (jug1,y),
            (x,jug2),
            (0,y),
            (x,0),
            (max(0,x-(jug2-y)),min(jug2,x+y)),
            (min(jug1,x+y),max(0,y-(jug1-x)))
        ]

        for state in next_states:
            if state not in visited:
                queue.append(state)
```

```
water_jug(4,3,2)
```

SAMPLE OUTPUT

```
(0,0)  
(4,0)  
(0,3)  
(4,3)  
(1,3)  
(3,0)  
(1,0)  
(0,1)  
(4,2)
```

Goal Reached

CONCLUSION

The Water Jug Problem was solved successfully by generating valid states until the desired quantity of water was obtained.